

GULSHAN KUMAR



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GitHub

Education

- National Institute of Technology (NIT) Agartala** CGPA: 8.39
B.Tech in Computational Mathematics 2024 – 2028
- Class XII (CBSE)** Percentage: 89%
Higher Secondary Education 2023
- Class X (CBSE)** Percentage: 92.14%
Secondary Education 2021

Projects

- Amazon Home Page Clone** Nov 2025
Responsive E-Commerce Front End [Live Demo Link](#)
 - Frontend Engineering:** Developed a high-fidelity replica of the Amazon homepage using **Semantic HTML5** and **CSS3**, replicating the complex multi-column layout and visual hierarchy.
 - Layout Design:** Implemented advanced **CSS Flexbox** and Box Model techniques to structure the navigation bar, hero section, and dynamic product grids without relying on external frameworks.
 - Cross-Platform Optimization:** Engineered a fully **responsive interface** that adapts to varying screen resolutions, ensuring consistent content alignment and readability.
 - Tech Stack:** HTML5, CSS3, Git, Javascript, Version Control.
- DFS Path Finder** Dec 2025
Interactive Graph Traversal Engine [Live Demo Link](#)
 - Algorithm Implementation:** Developed a traversal engine utilizing **Depth-First Search (DFS)** to compute path connectivity in complex **unweighted graph topologies**.
 - Data Structure Logic:** Implemented an iterative **Stack (LIFO)** mechanism to simulate recursive backtracking, ensuring robust path exploration with **Time Complexity** $O(V + E)$.
 - Dynamic Visualization:** Engineered a responsive grid system that renders **real-time backtracking** and node state changes using asynchronous DOM manipulation.
 - Tech Stack:** JavaScript (**ES6+**), HTML5, CSS3, Data Structures (Stacks, Graphs).
- News Navigator** Mar 2026
AI-Powered Intelligence Feed [GitHub Link](#)
 - AI & System Architecture:** Engineered a decoupled pipeline using **Python (Flask)** and **Google Gemini 2.5 Flash** to autonomously extract financial entities, analyze market sentiment, and generate vernacular translations.
 - API Integration & Resiliency:** Integrated **NewsAPI** via **RESTful APIs** for real-time data ingestion, implementing strict JSON-mode prompt engineering and robust error-handling logic to prevent UI crashes.
 - State Management & UI:** Developed a responsive frontend using **JavaScript** and **Tailwind CSS**, utilizing a Memory Manager with **localStorage** to track user affinities and dynamically retune personalized news feeds.
 - Tech Stack:** Python, Flask, JavaScript, REST APIs, Google GenAI SDK, NewsAPI.

Achievements

- Ideathon Winner:** Secured a **Top 3 Rank** in an inter-college Ideathon at NIT Agartala, competing against teams from multiple institutions and earning a cash prize for innovative problem-solving.
- Tech Fellowship:** Secured **2nd Position** as an **ATF Scholar 2025** in the **AlgoUniversity Tech Fellowship (ATF)** selection process among candidates at NIT Agartala. [\[View Certificate & Award\]](#)
- Algorithmic Excellence:** Rated **3-Star Coder** at CodeChef ([gulshan3728k](#)) demonstrating consistent problem-solving under time constraints.
- Global Problem Solving:** Solved **400+ Algorithmic Problems** across LeetCode and Codechef, focusing on Dynamic Programming and Graph Theory. [\[LeetCode Profile\]](#)
- Hackathon Innovation:** Qualified for the Internal Round of **Smart India Hackathon (SIH) 2025**, selected for proposing a novel technical solution to a real-world problem statement.

Technical Skills

- Languages:** C, C++, JavaScript, Python, SQL, HTML5, CSS3
- Backend & Frameworks:** Node.js, Express.js, Flask, REST APIs, RESTful API Design
- Databases:** MySQL, MongoDB, Database Design, CRUD Operations
- Core Competencies:** Data Structures and Algorithms (DSA), Object-Oriented Programming (OOPs), System Design Basics, Time and Space Complexity Analysis
- Developer Tools:** Git, GitHub, VS Code, Postman, Chrome DevTools
- Soft Skills:** Agile Methodology, Technical Documentation, Collaborative Problem Solving